

Sujin(Joanna) Yoo

MASTER'S STUDENT · APPAREL DESIGN / COMPUTER SCIENCE

Ithaca, NY

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Education

Cornell University

Ithaca, NY

M.A. APPAREL DESIGN WITH A MINOR IN COMPUTER SCIENCE

Aug.2024 - May.2026 (Expected)

- Advisors: Dr. Fatma Baytar (chair), Dr. Steve Marschner (minor committee)

University of Southern California (USC)

Los Angeles, CA

B.S. COMPUTER SCIENCE AND BUSINESS ADMINISTRATION

Aug.2021 - May.2024

- Combined Bachelor of Science degree program

Los Angeles Trade-Technical College

Los Angeles, CA

A.S. MATHEMATICS

Jan.2019 - May.2021

- Earned concurrently with full-time employment

Fashion Institute of Design and Merchandising (FIDM)

Los Angeles, CA

A.A. FASHION DESIGN

Mar.2015 - Jun.2017

Research Projects

Layer-Based Garment Modeling for Textile Form Weaving

Inria Centre CRAFT Team

ADVISED BY DR. MELINA SKOURAS AND DR. ADRIEN BOUSSEAU

Incoming, Starting Feb 2026

- Will join the INRIA CRAFT team in Spring 2026 to develop multilayer textile-form weaving simulation tools, including mathematical constraint modeling and interactive visualization

Hyperbolic Tessellation for Apparel Pattern Construction

Cornell University

ADVISED BY DR. YUSHENG LUO

Ongoing

- Developing an apparel pattern model based on hyperbolic tessellation, using the $\{5, 4\}$ pentagonal tiling as a generative structure.

Recommendation System for Maximizing Scrap-Fabric Utilization in Home Sewing

Cornell University

ADVISED BY DR. FATMA BAYTAR AND DR. STEVE MARSCHNER (INDEPENDENT PROJECT)

Ongoing

- Designing a recommendation system that analyzes scrap fabric images to suggest projects and maximize creative use for home sewers

Interactive Size-Chart and Automated Pattern Adjustment System for Home Sewers

Cornell University

ADVISED BY DR. FATMA BAYTAR AND DR. STEVE MARSCHNER (MASTER'S THESIS)

Ongoing

- Developing a 3D interactive size-chart and automated pattern adjustment tool integrated with CLO3D to help home sewers with non-standard body types
- Awarded Alan D. Mathios Research and Service Grant, Cornell University (\$1,000)
- 1st Place – HCD Graduate Student Poster Contest, Cornell University (\$250)

Posture Descriptors (PARCS): Improving Posture Analysis for Garment Fit

Cornell University

CORNELL FASHION & BODY TECH LAB

2025

- Assisted with 3D body scanning, pose estimation, and data preparation

Developing Inclusive Swim Cap Design Using 3D Body Scanning

CORNELL FASHION & BODY TECH LAB

Cornell University

2024

- Assisted with 3D body scanning and used Geomagic software to clean, process, and analyze head-shape data

Teaching Experience

- 2025 **DEA 1140: Principles of Design Computing**, Graduate Student Lecturer, Cornell University
- 2025 **FSAD 1450: Introduction to Fashion Design**, Teaching Assistant, Cornell University
- 2024 **FSAD 1450: Introduction to Fashion Design**, Teaching Assistant, Cornell University
- 2024 **Integration of Artificial Intelligence in the Fashion Industry**, Guest Lecturer, StyleCAD USA
- 2022-2024 **Software Programming Instructor**, iANT Education, Los Angeles, CA

Professional Experience

- 2024-ing **Creative Director**, Joanna Blüm Studio
- 2022-2024 **Freelance Technical Designer**, Beautiful Day Wedding
- 2023 **Freelance Technical Designer**, Brochu Walker
- 2021-2022 **Freelance Senior Fashion Designer**, Ginger Green
- 2021 **Senior Fashion Designer**, Mystree
- 2018-2021 **Associate Fashion Designer**, Papermoon Clothing
- 2017-2018 **Associate Fashion Designer**, Oh Yes Fashion
- 2016-2017 **Assistant Fashion Designer**, Aime
- 2015-2016 **Graphic/Apparel Designer**, Young 2 Dress

Publications

Yoo, S.; Lopez, M.; Marschner, S.; Baytar, F. Designing an Interactive Size Chart to Improve Fit in DIY Garment Making. *International Conference on Human-Computer Interaction (HCI) 2026*, accepted.

Baytar, F., Yang, Y., Maher, M., Rivera, I., **Yoo, S.**, Patel, S. (2025). Implementing 3D body scanning methods for inclusive swim cap design and sizing. *HCI 2025*, Springer LNCS 15792. https://doi.org/10.1007/978-3-031-93505-3_1

McDonald, C.; Navodhya, P. U.; Ruderman, G.; Carmicino, B.; Baytar, F.; Maher, M.; **Yoo, S.** (2025). Posture Descriptors (PARCS): What We Know and the Gaps. *Proceedings of 3DBODY.TECH 2025*. doi:10.15221/25.37

UNDER REVIEW:

Yoo, S.; Baytar, F. Fabric Scraps as Transitional Materialities: A Thematic Analysis of Sewing and Craft Practices. *Design Research Society (DRS) 2026*, under review.

Awards, Fellowships, & Grants

- 2025 **Recipient, Oris Glisson Fellowship**, International Textile & Apparel Association \$ 1,000
- 2025 **1st Place, HCD Graduate Student Poster Contest**, Cornell University \$ 250
- 2024 **Recipient, Alan D. Mathios Research and Service Grant**, Cornell University \$ 1,000
- 2023 **Outstanding Team Project Award (Android App)**, University of Southern California

Relevant Coursework

Cornell University:

CS5620 Intro to Computer Graphics, **CS5643** Physical Based Animation for Computer Graphics, **CS5621** Computer Graphics Practicum, **MATH4520** Classical Geometries and Modern Applications, **FSAD6025** Design for Change, **FSAD3650** New Technologies for Fashion Design

Other Institutions:

- Completed full lower-division mathematics sequence (Calculus I–III, Linear Algebra, Differential Equations)
- Completed core undergraduate computer science courses (Data Structures, Algorithms, Discrete Mathematics, Artificial Intelligence, etc.)

Skills

- C++, C#, Python, Java, TaichiLang, PyQt, Unity, OpenGL, OpenCV, Blender, Geomagic, Qt, CLO3D, Browzwear, Illustrator, Latex, GitHub
- Amazon Web Services(AWS) Certified Cloud Practitioner, Certificate Verification Number: CT40GH217J44Q1KF
- Oracle Certified Associate, Java SE 8 Programmer, Certificate Verification Number: 20425079OCAJSE8